

Definition: set

A set is a collection of objects (referred to as elements or members).

Sets are written using $\{$ and $\}$

e. g. Set of positive numbers = $\{2, 4, 6\}$

$A = \{1, 2, 3\}$ and $B = \{2, 3\}$

The order of the elements themselves do not really matter.

Definition: Cardinality

The number of elements

$\in$ = 'in' or 'is an element'

Normally, we use set constructor notation to form a new set from an existing one.

Set of positive integers =
 $\{1, 2, 3, \dots\} = \mathbb{Z}^+$

Set of positive integers below 10

$= B = \{n \in A : n < 3\} = \{1, 2\}$

$\emptyset$ = Empty set / set with no elements

$\mathbb{N}$ = Natural numbers = $\{1, 2, 3, \dots\}$

$\mathbb{Z}$ = Integers = $\{\dots, -1, 0, 1, \dots\}$

$\mathbb{Q}$ = Rationals = $\left\{ \frac{n}{m} : m, n \in \mathbb{Z} \text{ with } m \neq 0 \right\}$

$\mathbb{R}$ = Real numbers

$\in \mathbb{R}$

Definition: Subset

Let A be a set. A subset of A is a set B such that every element of B is an element of A .

' B is a subset of A ' is written as $B \subseteq A$ or $B \subset A$

For any set A , $A \subseteq A$ and $\emptyset \subseteq A$

$A = \{1, 2, 3\} \Rightarrow 8$ possible subset

$\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$
and $\{1, 2, 3\}$
All ordered

Number of subsets = 2^n

Definition: Equality of sets

Let A and B be sets. Then $A = B$ if
 $A \subseteq B$ and $B \subseteq A$

$A \cup B$ = union of A and B
 $= \{x : x \in A \text{ or } x \in B\}$

$A \cap B$ = intersection of A and B

$$= \{x : x \in A \text{ and } x \in B\}$$

$$A \setminus B = \text{set-theoretic difference} \\ = \{x \in A : x \notin B\}$$

A powerset is the set of all

$$[|S| = n] \Leftrightarrow [P(S) = 2^n]$$

$$A \times B = \text{Set of ordered pairs } (a, b) \\ \text{with } a \in A \text{ and } b \in B$$

$(a, b) \neq (b, a)$ as the order is different

$A \times B$ is known as the cartesian product (or just product of A and B).

$$A = \{1, 2, 3\} \quad , \quad B = \{a, b\}$$

$$A \times B = \{(1, a), (2, a), (3, a), (1, b), (2, b), (3, b)\}$$

$$A^2 = A \times A \Rightarrow A^n = A \times A \times \dots \times A \quad n \text{ Times}$$

$$\text{If } |A|, |B| = 'm, n', \text{ then } |A \times B| = mn$$